Mirror Principle and Split VoiceP

Evidence from Mongolian and Japanese[1]

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This paper demonstrates that the layered affixation of passive and causative morphemes in Mongolian and Japanese mirrors the Split VoiceP structure of passive and causative clauses, where the affixation of a voice suffix signals the introduction of an argument by a voice head spelled out by that suffix. This head-argument correlation, ensuring the morpho-syntax mirroring of voice suffixes, entails that the voice domain comprises separate projections that are split out of VoiceP. Clauses are built by introducing arguments as potential subjects (sbjs) through voice heads of the same substance, where a last-introduced sbj is promoted to the nominative position, with others, if any, remaining non-nominative within the voice domain. From this, it follows that Voice is a sbj-introducing, not merely argument-introducing, head, and that the interconnection of passives and causatives as voice constructions lies in the fact that both are derived over the Split VoiceP structure. Consequently, voice constructions can be accounted for in a unified way, without the need of postulating dedicated heads such as Passive and Cause, and the clause building mode is reducible to the introduction of arguments as sbjs, which ultimately comes down to Free Merge.

Keywords Mirror Principle, passive, causative, voice, Mongolian, Japanese

1. Introduction

This paper addresses two primary questions, one concerning the syntactic properties of causative-passive interactions in two agglutinating languages, Mongolian and Japanese, and the other concerning the implications they carry for the current generative theory of building clauses. Explicating these issues refers us to Mirror Principle, first advocated by Baker (1985), who formulates that morphological operations, including linear ordering of words and affixes, and syntactic operations mirror each other. The mirroring between hierarchical syntactic and morphological structures extends to their semantics, producing corresponding meanings scopally interpreted for the structures. In (1), for example, the causative word *made*, merged later than the desiderative *want*, has a different interpretation than (2), in which the ordering of causative and desiderative is reversed, mirroring their reverse merging order in the hierarchical syntactic derivation.

(1) John made Bill want to go.

(2) John wanted to make Bill go.

Where the causative-passive interaction is concerned, increasing and reducing argument roles and affixing morphemes are correlated as a Mirror Principle effect. Assume that Taro is the

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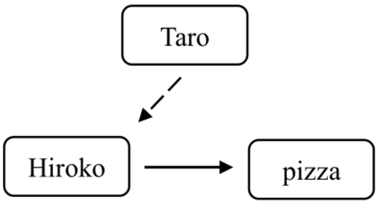
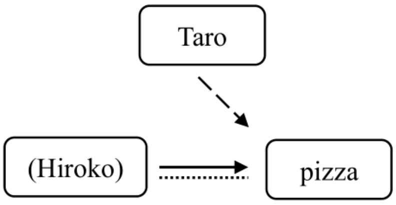
causer, Hiroko is the agent and the pizza is the patient. At least three distinct types of affixing ordering between causative and passive morphemes are then available, according to whether the agent or the patient is presented as the causee and whether the causer or the agent is presented as the subject. Type 1 differs from Types 2 and 3 in that it has no passivizing relation. Types 2 and 3 both have a specific passivizing relation. In Type 2, passivization involves the agent and the patient, whereas in Type 3, it involves the agent and the causer. Type 2 differs from Types 1 and 3 in that the causation relationship takes place between the causer and the patient.[2]

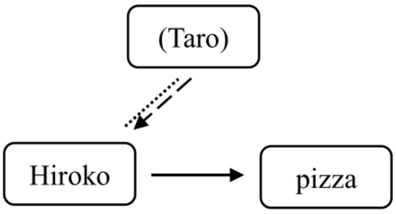
(3) Type 1: Subject (Taro), Causee (Hiroko), Object (pizza)
Intended meaning: Taro causes Hiroko to eat the pizza.



Type 2: Subject (Taro), Causee (pizza) (Hiroko, the agent, is suppressed.)
Intended meaning: Taro causes the pizza to be eaten (by Hiroko).



Type 3: Subject/Causee (Hiroko), Object (pizza) (Taro, the causer, is suppressed.)
Intended meaning: Hiroko is caused to eat the pizza (by Taro).



It is then intriguing to question how languages differ in terms of the implementation of Mirror Principle that is visualized by affixing orderings and/or word order. In non-agglutinating languages such as Chinese and English, individual words are employed in spelling out the arguments and the relationships among them. In contrast, in agglutinating languages such as Mongolian and Japanese, the relations among the arguments are realized as individual words created out of separate affixes, each of which represents a certain terminal node in the hierarchical structure. For example, in (4), *-sase*, coordinating with *Taroo*, renders causation,

[2] The solid lines, dashed lines and dotted lines represent transitivity, causation and passivization, respectively. The arrows represent the ordering of the two related arguments linked by each arrowed line.

with the agent *Hiroko* ending up as a non-nominative (NOM) element. When *-rare* is attached to *tabe-sase*, it calls for a NOM element, which induces passivization of *Hiroko* to become that NOM element, as in (5). Causativization followed by passivization is thus spelled out by *-sase* embedded under *-rare*, with *Taroo* introduced first and subsequently suppressed by promoting *Hiroko* as a candidate for the NOM position. Similarly, in (6), the affixation of *-rare* goes with the reintroduction of the patient *Hanako* as the subject, instantiating passivization. In (7), the affixation of *-sase* goes with the introduction of *Ziroo* as the subject, instantiating causativization. Thus, the affixation of the morphemes *-are* and *-sase* in a given order as they occur in a word mirrors the consecutive occurrence of passivization and causativization in a corresponding order.

(4) *Taroo-ga* *Hiroko-ni* *pizza-o* *tabe-**sase**-ta*.[3]
Taro-NOM Hiroko-DAT pizza-ACC eat-CS-PST
'Taro made Hiroko eat the pizza.' (Harley 2008: 7-8)

(5) *Hiroko-ga* *pizza-o* *tabe-**sase-rare**-ta*.
Hiroko-NOM pizza-ACC eat-CS-PS-PST
'Hiroko was made to eat the pizza.' (Harley 2013: 53)

(6) *Hanako-ga* *Taroo-ni* *sikar-**are**-ta*.
Hanako-NOM Taro-DAT scold-PS-PST
'Hanako was scolded by Taro.' (Tsujimura 1996: 258)

(7) *Ziroo-ga* *Hanako-o/ni* *Taroo-ni* *sikar-**are-sase**-ta*.
Ziro-NOM Hanako-ACC/DAT Taro-DAT scold-PS-CS-PST
'Ziro made Hanako be scolded by Taro.' (Tsujimura 1996: 259)

It is not difficult to see that every time a dedicated morpheme is affixed, a (candidate for) subject occurs in a higher position, as informally represented in (8). In causativization, a new argument is introduced as a subject (via external merge/EM); in passivization, an old argument is reintroduced (via internal merge/IM).

(8) Head-argument correlation in passive-of-causative

	hierarchically low ↔ hierarchically high			
Root	patient			
transitive: ***tabe***	patient	**agent (S)**		
causative: *tabe-**sase***	patient	agent	**causer (S)**	
passive: *tabe-sase-**rare***	patient	*t*	causer	**agent (S)**

(9) Head-argument correlation in causative-of-passive

	hierarchically low ↔ hierarchically high

[3] The following abbreviations are used in this paper: 1PL (first-person plural suffix), 1SG (first-person singular suffix), ACC (accusative case suffix), CS (causative suffix), DAT (dative case suffix), GEN (genitive case suffix), INF (infinitive suffix), NEG (negation marker), NOM (nominative case suffix), NPST (non-past tense suffix), PS (passive suffix), PRS (present tense suffix), PRT (particle), PSS[3] (third-person possessive particle), PST (past tense suffix), RX (reflexive possessive particle), TOP (topic particle).

Root	patient			
transitive: ***sikar***	patient	**agent (S)**		
passive: *sikar-**are***	*t*	agent	**patient (S)**	
causative: *sikar-are-**sase***	*t*	agent	patient	**causer (S)**

It is thus quite clear that there is a one-to-one correlation between a verbal morpheme and a nominal argument, new or old, which is a candidate for the subject. Given that a verbal morpheme is the spell-out form of a syntactic head and the subject originates as an argument, the one-to-one correlation holds between a head and a subject argument introduced by it. On the other hand, mirroring holds between the affixation of verbal morphemes and (re)introduction of arguments, instantiating the morpho-syntax mirroring. Notably, this morpho-syntax mirroring is ensured by the head-argument correlation.

Relevant issues, presented around the head-argument correlation and the morpho-syntax mirroring, will be explained under the Split VoiceP approach, which is presented as a theoretical framework of syntactic Voice in this paper. Discussion is laid out around the question of how the head-argument correlation and the morpho-syntax mirroring are calculated in syntax. In figuring out this question, it is most important to identify the head that introduces the subject of passive clauses as well as that of active clauses.

In what follows, we first overview the basics of causatives and passives in Mongolian and Japanese (Section 2) and discuss how the interaction between causative and passive suffixes in the two languages mirrors that between causativization and passivization as syntactic operations (Section 3). After doing so, we will discuss how the Split VoiceP approach can capture the facts to be presented in Sections 2 and 3 (Section 4).

2. The empirical base: The passive-causative interaction

This section overviews the basic properties of passive and causative morphemes in Mongolian and Japanese, focusing on their interaction in a consecutive affixation.

2.1 Mongolian

The passive suffix (PS) in Mongolian *-GD* has three variants, as shown in (10).[4] Causative suffixes (CSs) in Mongolian include three categories *-UUL*, *-LGA*, and *-AA*, each with allophonic variants, complying with vowel harmony.

(10) PS and CS in Mongolian[5]

PS	*-GD: -gd, -d, -t*		
CS	*-UUL: -uul/üül, -guul/güül*	*-LGA: -lga/lge/lgo/lgö*	*-AA: -aa/ee/oo/öö, -ga/-ge/-go/-gö*

Among the passive variants, *-gd* is productive, and the other two are available for a very limited

[4] The three variants of *-GD* have Uyghur orthographical alternative forms *-gda/gde*, *-da/de* and *-ta/te*, respectively, which end with a short vowel, complying with vowel harmony. The classical orthography, also known as "the Uyghur-Mongolian script", is mainly used by speakers in China and the Cyrillic orthography is used in Mongolia.

[5] The Latin transliteration of Mongolian passive and causative suffixes in this paper is primarily based on the Cyrillic orthography. See Kullmann and Tserenpil (2015: 117ff) among others for the details of the orthographical rules about the causative suffixes.

number of verbs. The causative suffixes are all productive.

A salient property of these suffixes is that their ordering in the same verb has much to do with their functions and interpretations. CS either precedes or follows PS, as in (11-12) and (13-14). Importantly, the succession of two CSs, exemplified by (15-16), is quite productive, whereas the successive presence of two PSs, exemplified by (17), is extremely limited.

(11) *Itgegčid bolon tede-ne ariun sudaruud gal-d šat-**aa**-**gd**-b.*
followers and their holy books-NOM fire-DAT burn-CS-PS-PST
'The followers and their holy books were burned in the fire.' (*Alma* (Section Four))[6]

(12) *End olon üildber bai-**guul**-**gd**-ž bai-na.*
here many factory be-CS-PS-CV be-PRS
'Many facotiries are being established here.' (Kullmann and Tserenpil 2015: 130)

(13) *Hairlah sedgel ni uul yabdal-ig dahin dahin bod-**gd**-**uul**-na.*
loving heart TOP old thing-ACC again and again think-PS-CS-PRS
'The loving heart reminds me of old things repeatedly.' (*Tungalag Tamir* (Chapter One))[7]

(14) *Dorž-ig aab-d-ni tani-**gd**-**uul**-h-güi-in*
Dorž-ACC father-DAT-PSS[3] recognize-PS-CS-INF-NEG-GEN
tuld sahal naa-san.
for beard attach-PST
'In order not to make Dorž recognized by his father, I attached beard to his face.'
(Umetani 2006: 95)

(15) *Bi düü-geer hubčas-(ig)-aa hat-**aa**-**lga**-san.*
1SG-NOM younger sister-INS cloth-ACC-RX dry-CS-CS-PST
'I had my younger sister dried my clothes.' (Kullmann and Tserenpil 2015: 117)

(16) *Bagš nad-aar neg huudas züil orč-**uul**-**uul**-b.*
Teacher-NOM 1SG-SG-INS one page thing translate-CS-CS-PST
'The teacher had me translate one-page paper.' (Kullmann and Tserenpil 2015: 122)

(17) *Hutgan hürd zööltei yih zaan č usan-d ab-**ta**-**gd**-b.*
The wheeled big elephant-NOM even water-DAT take-PS-PS-PST
'Even the wheeled big elephant was carried away by flood.'
(*Shidet Khüür-in Ülger* (Chapter Two))[8]

PS-PS verbs include *ab-ta-gd*, *ol-da-gd*, *huur-da-gd*, etc. These verbs are morphologically bi-passives (passives of passive). However, they are interpreted as mono-passives (passives of

[6] This is a chapter of *Mormon-ii Nom: Yesüs Krist-iin Öör Negen Geree*, the Mongolian translation of *Book of Mormon: Another Testament of Jesus Christ*.
[7] This is a novel by the famous writer Chadraabalyn Lodoidamba from Mongolia.
[8] This is a folktale of Indian origin, known as "The Tale of the Magic Corpse", which was later translated to Mongolian.

active) and sometimes not even as passives. The fact that they alternate with *ab-t*, *ol-d* and *huur-d* indicates that they lack a bi-passive interpretation and function. All this prevents us from including PS-PS verbs in our target verb list. We thus have three types of suffixal orderings in Mongolian, with PS-PS being unavailable.

(18) a. CS-PS (see (11) and (12))
b. PS-CS (see (13) and (14))
c. CS-CS (see (15) and (16))

A few remarks are in order on the distinction and correlation between what is called 'causative' and what is called 'passive' here. It is noted that some verbs with a causative suffix CS can have a passive interpretation, as discussed by Washio (1993). Typical examples of such verbs include *haž* 'bite', *čohi* 'hit', *zanči* 'beat', *gišg* 'step on', *bari* 'catch, arrest' and *ab* 'take, steal'. However, these verbs, when affixed by a CS, do not express a purely passive meaning. Structurally, they are still causatives.[9] This is because the subject is still interpreted as being responsible for the event's happening, and therefore is a causer. For example, (19) is interpreted as "I got my hand bitten by the dog", nor "I was bitten by the dog on my hand", neither "My hand was bitten by the dog". Similarly, (20) is best translated as "I got (myself) bitten by the dog", not "I was bitten by the dog". It is (21) that means "I was bitten by the dog". Even if one wants to label weak causatives as passive, they are not canonical passives. The contrast between weak causatives with a passive meaning (V-CS) and canonical passives (V-PS) in Mongolian resembles that between *get*-passives and *be*-passives in English. "Passive" in this paper thus refers to a passive suffix PS or a relevant verb/sentence, and "causative" refers to a causative suffix CS or a relevant verb/sentence, when we discuss Mongolian.

(19) *Bi* *gar-(ig)-aa* *nohai-d* *haž-**uul**-san.*
I hand-ACC-RX dog-DAT bite-CS-PST
'I got my hand bitten by the dog.'

(20) *Bi* *nohai-d* *haž-**uul**-san.*
I dog-DAT bite-CS-PST
'I got (myself) bitten by the dog.'

(21) *Bi* *nohai-d* *haž-**gd**-san.*
I dog-DAT bite-PS-PST
'I was bitten by the dog.'

In Mongolian, a head-final language, the suffix that precedes the other is embedded under it hierarchically. An embedded suffix displays either a lexical or a syntactic property, whereas an embedding suffix only displays a syntactic property.[10] A passive/causative suffix with a lexical property is notated as "PS/CS_{lex}" and one with a syntactic property as "PS/CS_{syn}".

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[9] Such causatives are less preferred in the eastern dialects of Inner Mongolia than in other dialects.
[10] The lexical-syntactic distinction here does not have a theoretical implication. I am not arguing that "lexical" suffixes are attached to the root element in the lexicon while "syntactic" suffixes are attached to a stem (or a root) in syntax.

The distinction between lexical and syntactic properties of PS is suggested by the following facts.

First, the idiosyncrasy of PSs in form is indicative of their idiosyncrasy in meaning; the more compact PS is, the weaker the passive meaning becomes.[11] A literal passive reading is less possible with the morpho-phonologically more compact (and hence idiosyncratic) forms *-d* and *-t*,[12] as exemplified below.

(22) *Nad-d* *neg sonin duu* *sons-**d**-b.*
1SG-DAT a strange sound-NOM hear-PS-PST
'To me, a strange sound was heard.'
*'A strange sound was heard by me.'

In contrast, the regular, longer form *-gd* can have a literal passive reading, as exemplified in (21), in addition to an idiosyncratic reading, as exemplified in (23).

(23) *Nad-d* *hüč* *nem-**gd**-b.*
1SG-DAT strength-NOM add-PS-PST
'For me, strength increased.'
*'Strength was increased by me.'

The idiosyncrasy in meaning of verbs such as *nem-gd* lies in the fact that the dative element, if any, does not have an agentive interpretation, which otherwise is required in passives, and that that element is topical and therefore mostly occurs before the nominative element, which is not true for passives. Moreover, the demotion of the case of the argument that would otherwise be an agent from NOM to DAT takes place between the root verb (V), which is transitive, and the affixed verb (V-$\mathrm{PS}_{\mathrm{lex}}$), which is intransitive, as in (23). In this sense, we may regard *-gd* here as a detransitivizing, not passivizing, morpheme. All this makes the verb (V-$\mathrm{PS}_{\mathrm{lex}}$) behave like an unaccusative verb rather than a passive verb. For example, *neme-gd* means "increase (intr)", not "be added"; *üre-gd* can mean "die (a heroic death)", not necessarily "be consumed"; *sons-d* denotes the spontaneous and non-volitional action of "being heard", although it may be translated as a passive verb in English. All this is not true for V-$\mathrm{PS}_{\mathrm{syn}}$, which always requires the dative argument (the second argument) to have an agentive interpretation, as in (21).

Second, an embedding PS (PS in a CS-PS string) does not tolerate idiosyncrasy in both meaning and form. For example, *šat-aa-gd* is interpreted as "be burned", where there is an implicit agent.[13] Unlike *neme-gd* 'increase (intr)' and *sons-d* 'be heard', *šat-aa-gd* is not deprived of the agentivity; *šat-aa* 'burn (tr)', the causative form of *šat* 'burn (intr)', involves the agentivity, which is not overtly spelled out in syntax. In this sense, it is *šat* rather than *šat-aa-gd* that is analogous to *neme-gd* and *sons-d*, denoting spontaneous and non-volitional actions.

[11] For detailed discussion on this, see Bao et al. (2025).
[12] Typical examples, few though, include the following: *sons-d* 'be heard', *ol-d* 'be found', *huur-d* 'be deceived', *ab-t* 'lose, be taken', *hür-t* 'receive, (Literally 'be reached')'.
[13] In Mongolian, the dative case suffix *-d* is used to mark recipients, beneficiaries, locatives, causes, etc. in addition to agents in passives. In (24), *gal* 'fire' is a cause or a locative.

(24) *Itgegčid bolon tede-ne ariun sudaruud gal-d šat-**aa-gd**-b.*
followers and their holy books-NOM fire-DAT burn-CS-PS-PST
'The followers and their holy books were burned in the fire.' (= (11))

Regarding the lexical and syntactic properties of CS, the following facts are indicative of their distinction.

First, scopal ambiguity is observed with CS$_{syn}$, but not with CS$_{lex}$, when the causative is modified by preverbal adverbials such as *dahinta* 'again'. In (25), for example, *dahinta* 'again' takes either a wide scope associated with *ir-üül* 'cause to come' or a narrow scope associated with *ir* 'come'.

(25) *Bi Baatar-ig dahinta ir-**üül**-b.*
1SG-NOM Badma-ACC again come-CS-PST
'I [asked/allowed Badma come again].'
'I asked/allowed Baatar [to come again].'

Such adverbials can also precede the causer, where ambiguity is lost.

(26) *Bi dahinta Baatar-ig ir-**üül**-b.*
1SG-NOM again Baatar-ACC come-CS-PST
'I [asked/allowed Badma to come again].'

Given that such adverbials may be interpreted as modifying either the event denoted by V or the event denoted by V-CS, there can be two events of different levels for causatives. This leads to the claim that V is a morpheme at the lexical level and CS is one at the syntactic level, allowing us to say that CS (*-UUL* and *-LGA*, not *-AA*) in examples such as (25–26) is a syntactic causative morpheme (CS$_{syn}$). Importantly, *-AA* causatives disallow scopal ambiguity with such adverbials. As exemplified by (27), the adverbial modification goes for V-CS, not for V. This said, CS (*-AA* and sometimes *-UUL* and *-LGA* as well) in examples such as (27) is a lexical causative suffix (CS$_{lex}$).

(27) *Bid daisan-ig dahinta sön-**öö**-b.*
1PL-NOM enemy-ACC deliberately die out-CS-PST
'We eliminated the enemy again.'
'*We caused the enemy to die out/be eliminated again.'

Second, *-AA*, which is restricted to verbs with a direct causation meaning, never follows any other CS or PS; that is, *-UUL* and *-LGA* as well as *-GD* are never embedded under *-AA*.[14]

[14] This restriction on *-AA* partially follows from phonological rules. Importantly, when the same root/R is open for both *-AA* and *-UUL*, R-*AA*, not R-*UUL*, is interpreted as expressing a forced causation. For example, *hür-ge* means "deliver, send" and *hür-uul* means "cause to reach"; the former produces an idiomatic meaning when combined with objects, while the latter does not. Janhunen (2012: 149) states that in general, the suffix *-AA* is restricted to a gradually diminishing number of lexicalized items, while *-UUL* retains a considerably greater degree of synchronic productivity.

(28) *hat-**aa**-**lga*** (see Kullmann and Tserenpil 2015: 123)
dry-CS-CS
'cause to dry'

(29) *šat-**aa**-**gd*** (see (11/24))
burn-CS-PS
'be burnt'

(30) **orč-**uul**-**aa***
*orč-**uul**-**uul*** (see Kullmann and Tserenpil 2015: 122)
be-CS-CS
'cause to establish'

(31) **tani-**gd**-**aa***
*tani-**gd**-**uul*** (see (14))
kill-PS-CS
'cause to be killed'

Given that morphemes at the syntactic level are always attached higher than those at the lexical level, *-AA* can only be CS_{lex}.

Third, transmission of energy is observed between the causer and the causee in the case of lexical causatives, whereas it is not necessarily observed in the case of syntactic causatives.[15]

Importantly, a CS_{syn} can embed another, yielding the string CS_{syn}-CS_{syn}, exemplified in (32), which, however, is not very productive. Four arguments, nominative, instrumental, dative and accusative, respectively, occur in such causatives. The instrumental argument is a causee for the highest causation (i.e., I caused Dorž to …) and a causer in the embedded causation (i.e., Dorž caused my son to …).

(32)

Bi	*Dorž-oor*	*hüü-d-ee*	*em-ig-ni*	*uu-**lga**-**uul**-laa.*
1SG-NOM	Dorž-INS	son-DAT-RX	medicine-ACC-PSS[3]	drink-CS-CS-PST

'I asked Dorž to help my son drink the medicine.' (Kullmann and Tserenpil 2015: 119)[16]

We thus have four patterns of lex-syn interactions between CS and PS.

(33) a. CS_{lex}-PS_{syn}
b. PS_{syn}-CS_{syn}
c. CS_{lex}-CS_{syn}
d. CS_{syn}-CS_{syn}

PS_{lex} is not considered here because it, having been degrammaticalized into the verb stem, has

[15] See Hashimoto (2019) for detailed discussion of transmission of energy in Mongolian causatives.
[16] Kullmann and Tserenpil's (2015) translation of this sentence is "With Dorž's help, I made my son drink the medicine".

lost its passivizing function and should be considered a "fake" PS, which is still a lexical morpheme. Fake PS is exemplified by (13), (17), (22) and (23), in which the relevant morphemes, combined with the verb root, denote an inchoative change or spontaneous situation but not express a canonical, literal passive meaning, as noted earlier.

2.2 Japanese

There are two passive verb suffixes in Japanese, namely, *-are* and *-rare*, which are both productive. Common causative suffixes in Japanese include *-e*, *-s*, *-as*, *-ase* and *-sase*.

(34) PS and CS in Japanese

PS	*-are, -rare*
CS	*-e, -s, -os, -as, -ase, -sase*

Japanese has the same suffixal orderings as Mongolian (see (18)), lacking PS-PS.[17] CS precedes PS in (35–36) and follows it in (37–38). In (39–40), CS precedes/follows another CS.

(35) *Hoteru-no mae-de* *or-**os**-**are**-ta.*
in front of hotel drop-CS-PS-PST
'I was dropped off in front of the hotel.'

(36) *Hiroko-ga* *pizza-o* *tabe-**sase**-**rare**-ta.*
Hiroko-NOM pizza-ACC eat-CS-PS-PST
'Hiroko was made to eat the pizza.' (= (5))

(37) *Ziroo-ga* *Hanako-o/ni* *Taroo-ni* *sikar-**are**-**sase**-ta.*
Ziro-NOM Hanako-ACC/DAT Taro-DAT scold-PS-CS-PST
'Ziro made Hanako be scolded by Taro.' (= (7))

(38) *Musuko-o* *seken-no aranami-ni* *mom-**are**-**sase**-ru.*
son-ACC world-GEN rough waves-DAT rub-PS-CS-PRS
'(I will) let my son be buffeted about in the hardships of the world.' (Washio 2018: 118)

(39) *Taroo-ga* *Jiroo-ni* *sensei-o* *nak-**as**-**ase**-ta.*
Taro-NOM Jiro-DAT teacher-ACC cry-CS-CS-PST
'Taro made Jiro make the teacher cry.' (Pylkkänen 2008: 122)

(40) *Reiko-ga* *Hanako-ni* *yoofuku-o* *aw-**ase**-**sase**-ta.*
Reiko-NOM Hanako-DAT clothing-ACC meet-CS-CS-PST
'Reiko made Hanako match her clothing.' (Miyagawa 1998: 88)

What concerns us for our present purposes is the lexical-syntactic distinction of PS as well as CS. While the lexical and syntactic properties of causatives have extensively been discussed, few studies have focused on those of passives.

[17] Aoyagi (2021: 99ff) presents certain examples of PS-CS that are not fully ungrammatical but highly awkward.

As noted above, PS in Japanese PS-CS seems to always be PS_{syn}, exemplified by (37) and (38). PS in CS-PS, too, is always PS_{syn}. This is to say that PS_{lex} (for lexical passive) does not exist in Japanese. Recall that Mongolian, unlike Japanese, has PS_{lex}, which is readily available (see (13), (22), and (23)). *Kuwae-rare* 'be added' and *kik-are* 'be heard', the corresponding Japanese forms of Mongolian *neme-gd* 'increase (intr)' and *sons-d* 'be heard', have retained a literal passive meaning. This is evidenced by the fact that the dative element is readily interpreted as an agent (or an experiencer) and normally it, being non-topical, does not precede the nominative element. In this sense, *-(r)are*, functioning properly as a passivizing suffix, is not PS_{lex}, which is not true of Mongolian *-GD* in the corresponding forms such as *neme-gd* and *sons-d*. *Omow-are*, literally meaning "be thought", the corresponding form of Mongolian *bod-gd*, is a tricky case. Despite its passive morphology, this verb does not express a canonical passive meaning "X be done by Y" in that the agent (or experiencer), when present, is nominative, functioning as a subject. Moreover, it allows an accusative object. In this sense, it functions as an active verb. Notably, however, case-demotion does not occur between *omow-are* (V-PS) and *omow* (V). This renders *omow-are* different from Mongolian *bod-gd*, in which the case of the agent (or expierencer) argument is demoted from NOM to DAT. Accordingly, *-are* in *omow-are* is neither PS_{syn} nor PS_{lex}. In other words, it is not a passivizing morpheme in the stricter sense. Its proper function is just to "delete the [volition] feature from the action" to make it non-volitional.

To sum up, PS in the examined Japanese forms corresponding to the Mongolian PS_{lex} verbs are not PS_{lex}. Consequently, Japanese lacks PS_{lex}, as far as we can tell from the above observation.

The lexical-syntactic distinction of CS can be tested by a few facts. According to Harley (2008), lexical causatives in Japanese are monoclausal, can have idiomatic interpretations, have allomorph, allow a strong speaker sense of 'listedness' and may feed (non-productive) nominalization. An overview of the details is left out here. Additionally, the relatively compact causative forms including *-e*, *-s*, *-os*, *-as*, and *-ase* do not follow the productive causative form *-sase* and the passive form *-(a)ase* and they can mostly exhibit the lexical properties summarized by Harley (2008). In contrast, a suffix, CS or PS, following another is not compact and does not exhibit those lexical properties. This, resembling the combination patterns of the Mongolian suffixes as in (28–31), points to the lexical-syntactic distinction of morphological causatives to a further extent.

For CS-PS, CS may be either CS_{lex} or CS_{syn}, exemplified by (35) and (36). Unlike Mongolian PS-CS, however, PS in Japanese PS-CS seems to always be PS_{syn}, as exemplified by (37) and (38). For CS-CS, the embedded CS in CS-CS is always CS_{lex}, and the embedding CS is always CS_{syn}. This makes Japanese different from Mongolian again.[18]

We thus have four patterns of lex-syn interactions between CS and PS in Japanese.

(41) a. CS_{lex}-PS_{syn}
b. PS_{syn}-CS_{syn}
c. CS_{lex}-CS_{syn}

[18] See also Kuroda (1993: 8-10), who observes that an embedded causative morpheme is a lexical one. Aoyagi (2021: 101), however, presents a fact that is uncomfortable with Kuroda's (1993) observation.

d. CS_{syn}-PS_{syn}

Unlike Mongolian causatives and passives, those in Japanese have been extensively discussed by linguists, especially by Chomskyan generativists, who are particularly concerned with their morpho-syntax interface and related properties. For discussion of the lexical-syntactic distinction of Japanese causative suffixes, see Shibatani (1976), Jacobsen (1992), Kuroda (1993), Miyagawa (1984, 1998) and Harley (2008), Pylkkänen (2008), among many others; for discussion of structural elaboration on causatives and passives, see Kubo (1992), Pylkkänen (2008), Harley (2013) and Aoyagi (2021), among many others; for discussion of causative-passive interactions, see Saito (1982), Tsujimura (1996) and Aoyagi (2021), among many others; and for comprehensive reviews of discussions and analyses of Japanese passives, see Shibatani (1990) and Hoshi (1999).

2.3 Interim summary

Summarizing from Section 2.1 and Section 2.2, a total of five patterns of causative-passive interaction are available in Japanese and Mongolian.

(42) Paradigmatic comparison of causative-passive interactions in Mongolian and Japanese

	Mongolian	Example	Japanese	Example
CS_{lex}-PS_{syn}	YES	(11)	YES	(35)
PS_{syn}-CS_{syn}	YES	(14)	YES	(37), (38)
CS_{lex}-CS_{syn}	YES	(15)	YES	(39), (40)
CS_{syn}-CS_{syn}	YES	(32)	NO	
CS_{syn}-PS_{syn}	NO		YES	(36)

A few notes are in order on (42). First, the first three patterns are available in both languages, where CS_{lex}-PS_{syn} and CS_{lex}-CS_{syn} are highly productive, whereas PS_{syn}-CS_{syn} is not.[19] Second, CS_{syn}-PS_{syn} is available in Japanese but not in Mongolian. However, what Mongolian has is not available in Japanese, namely, CS_{syn}-CS_{syn}. Third, notably, neither language allows lex to follow syn.

3 Affixing order complying with Mirror Principle

Given that affixation as a morphological operation is the realization of a syntactic operation, there are at least *three distinct levels* of syntactic operations for the five patterns summarized in (42), in accordance with Mirror Principle. The lowest level of operation is correlated with CS_{lex} and the other two levels are correlated with the successive occurrance of CS_{syn}/PS_{syn} and PS_{syn}/CS_{syn}. When a single CS_{syn} or PS_{syn} occurs, it is correlated with the second high level of

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[19] Sentences with the pattern of PS_{syn}-CS_{syn}, instantiating what can be called "causative-of-passive", are not highly productive arguably due to the blocking by an alternative structure that can express a causative-of-passive meaning. In that alternative, PS is absent but a passive meaning is expressed. I label it "causative-with-passive-meaning" (e.g., *bagš-d zangn-uul* 'Lit. get oneself criticized by the teacher'). The causative with passive meaning is not unique to Mongolian; it is rather a highly general phenomenon cross-linguistically, being observed, in particular, in Korean, French, and Japanese in addition to Mongolian (Washio 1993: 46ff). It is also found in Chichewa and many other Bantu languages (Alsina 1992: 518ff and the sources therein).

operation. Since a syntactic operation does not feed a lexical operation in verb-formation, the affixing order of lex1, syn2 and syn3 will look like (43) linearly, with a lexical suffix preceding a syntactic suffix.

(43) [$_{XP3}$ [$_{XP2}$ [$_{XP1}$ […] -lex1] -syn2] -syn3]

Each of lex1, syn2 and syn3, spelling out the head of XP1, XP2 and XP3, respectively, corresponds to a distinct type of syntactic operation, causativization or passivization. The two operations are not differentiated for the moment.

The layered structure of [$_{XP2}$ [$_{XP1}$ […]]], mirroring the -lex1-syn2 string, has been widely discussed in the literature, including Pylkkänen (2008), Harley (2013), Wurmbrand and Shimamura (2017), Nie (2020) and many others, whereas the implementation of [$_{XP3}$ [$_{XP2}$ […]]], which mirrors -syn2-syn3, has not been explicated. The structure [$_{XP2}$ [$_{XP1}$ […]]] is readily exemplified by (11), (15), (35), (39) and (40). As shown by these examples, lex1 (*-AA* in Mongolian and *-as(e)* in Japanese) is the exponent of a lexical-causativizing head, CS_{lex} here, which has sometimes been assumed to be a transitivizing head, v or Voice, in the literature. What is spelled out by syn2 is a causativizing or passivizing head, CS_{syn} or PS_{syn}.

In (44), a passive sentence, the promotion of the subject *Dorž* is associated with affixation of PS *-gd*. That is, passivization as a syntactic operation mirrors the affixation of a passive suffix. In (45), which represents what I label "causative-of-passive" in this paper, when the causer argument, the subject *bi*, enters the derivation, the affixation of CS *-uul* takes place. Note that before the causer is introduced, the patient *Dorž* is in-situ, linearly following the agent *aab* and preceding the verb. In (45), *Dorž* undergoes scrambling and precedes *aab*. Importantly, the affixation of *-gd* is bound with the introduction of *Dorž*, and the affixation of *-uul* is bound with the introduction of the causer *bi*.

(44) *Dorž* *aab-d-aa* *tani-**gd**-san.*
Dorž-NOM father-DAT-RX recognize-PS-PST
'Dorž was recognized by his father.'

[*Dorž* [*aab* [~~*Dorž*~~ *tani*] ***-gd***] ...[20]

(45) *Bi* *Dorž-ig* *aab-d-ni* *tani-**gd**-**uul**-h-güi-in*
1SG-NOM Dorž-ACC father-DAT-PSS[3] recognize-PS-CS-INF-NEG-GEN
tuld sahal naa-san. (adabted from (14))
for beard attach-PST
'In order not to make Dorž recognized by his father, I attached beard to his face.'

[*Bi* [*Dorž* [*aab* [~~*Dorž*~~ *tani*] ***-gd***] ***-uul***] ...[21]

Because the affixation of *-gd*, a passive suffix, precedes that of *-uul*, a causative suffix, the promotion of *Dorž*, the patient, precedes the introduction of *bi*, the causer. Under the traditional

[20] The agent, being the highest argument, is promoted to SBJ.
[21] The causer, being the highest argument, is promoted to SBJ.

generative view, it would be the case that the promotion of *Dorž* is not bound with and is quite later than the affixation of *-gd* because a patient promotion is assumed to have nothing to do with a passive marker on the verb. From the perspective of Mirror Principle, however, a passive suffix, at least in agglutinative languages such as Mongolian and Japanese, is the very exponent of the head that induces passivization. That head cannot be the locus of tense T, although the patient may end up in Spec of TP in the surface structure. Japanese causative-of-passive sentences are more explicit about this point.[22] Unlike Mongolian causative-of-passive sentences, in which the passivized patient can only be assigned ACC, Japanese causative-of-passive sentences provide two options of case to be assigned to the passivized patient.

In (47), the patient *Hanako* occurs with DAT, indicating that it has moved away from its base-generated position, where it would otherwise be assigned ACC, to a position lower than Spec of T, a NOM position.[23] As indicated by PS *-are* in (46) and (47), what *Hanako* undergoes is passivization, followed by causativization in (47). In both (46) and (47), *Hanako* cannot be base-generated in its surface position because its surface position is not thematic; the passivizing head, spelled out as *-are*, is not a theta-role assigner. I label this type of movement "A-to-D raising" (raising from ACC to DAT). A-to-D raising does not apply to Mongolian, as illustrated in (45), in which the patient *Dorž* is obligatorily ACC.[24]

(46) *Hanako-ga* *Taroo-ni* *sikar-**are**-ta.*
Hanako-NOM Taro-DAT scold-PS-PST
'Hanako was scolded by Taro.' (= (6))

(47) *Ziroo-ga* *Hanako-o/ni* *Taroo-ni* *sikar-**are**-**sase**-ta.*
Ziro-NOM Hanako-ACC/DAT Taro-DAT scold-PS-CS-PST
'Ziro made Hanako be scolded by Taro.' (= (7/37))

[*Ziroo* [*Hanako* [*Taroo* [~~*Hanako*~~ *sikar*] ***-are***] ***-sase***] ...

Importantly, CS *-sase* in A-to-D raising is different from that in regular transitive-based causatives in that the height of its affixing position differs. In a regular transitive-based causative such as (48), *-sase*, selecting an active structure, is merged right above what has been assumed to be vP (corresponding to Voice1 in Section 4), whereas in A-to-D raising (47), it, selecting a passive phrase, is merged higher than what selects vP.

(48) *Ziroo-ga* *Taroo-ni* *Hanako-o* *sikar-**ase**-ta.*
Ziro-NOM Taro-DAT Hanako-ACC scold-CS-PST
'Ziro made Taro scold Hanako.'

[22] Various labels of this construction occurred in the literature. Tsujimura (1996) used "causative passive", Harley (2008) used "indirect causative", and Aoyagi (2021) used "causative-passive".

[23] See Saito (1982: 92), Hoshi (1999: 204), Aoyagi (2021: 99), etc., for relevant discussion on this construction.

[24] This asymmetry between Japanese and Mongolian has to do with the difference between the two languages in the parameter related to case-assignment of the cause and certain semantics-related parameters. In Japanese, the causee can either be ACC or be DAT in both transitive- and intransitive-based causatives, which is not true of Mongolian.

[*Ziroo* [*Taroo* [*Hanako* *sikar*] ***-ase***] …

Note that in (47), the causee is the patient *Hanako*, and in (48), it is the agent *Taroo*. Complying with Mirror Principle, since the causee is (re)merged right after the agent is merged, a corresponding affixation of a suffix takes place. In (47), PS *-rare* is affixed after *Taroo* is merged, whereas in (48) *-sase* is affixed in connection with the introduction of *Taroo*. Since the affixation of *-sase* follows that of *-rare* in A-to-D raising, *-sase* is merged higher in A-to-D raising than in regular causatives.

Again, mirroring between syntactic and morphological operations is evidenced by A-to-D raising in Japanese causative-of-passives. In sum, in Mongolian and Japanese causative-of-passive sentences, the passive morphemes *-GD* and *-rare* exemplify syn2, and the causative morphemes *-UUL* and *-sase* exemplify syn3.

We proceed to discuss Japanese passive-of-causative sentences (agent-promoting passives),[25] in which the order of CS and PS is reversed, with CS being syn2 and PS being syn3. What has been puzzling in causative-of-passive sentences is $\mathrm{PS}_{\mathrm{syn3}}$, which was often undifferentiated from $\mathrm{PS}_{\mathrm{syn2}}$ in direct passive sentences in previous studies. Morphology alone is not helpful in distinguishing them. Instead, we need to look at their syntactic properties. The following pair is compared.

(49) *Neko-ga* *John-ni* *nezumi-o* *tabe-**sase-rare**-ta.*
cat-NOM John-DAT mouse-ACC eat-CS-PS-PST
'The cat was made to eat the mouse by John.' (Aoyagi 2021: 88)

[*Neko* [*John* [*~~neko~~* *nezumi* *tabe*] ***-sase***] ***-rare***] …

(50) *Nezumi-ga* *neko-ni* *tabe-**rare**-ta.*
mouse-NOM cat-DAT eat-PS-PST
'The mouse was eaten by the cat.' (Aoyagi 2021: 87)

[*nezumi* [*neko* *~~nezumi~~* *tabe*] ***-rare***]

In both sentences, *nezumi* 'mouse' is the patient. When *nezumi* is promoted, PS *-rare* is attached to the root verb. In contrast, when *nezumi* stays in-situ, functioning as an object, *-rare*, in fact, is not attached to the root, as seen in (49). Therefore, in this case, *-rare* is not associated with the patient *nezumi* but with the promoted agent *neko*. Now, in (49) the affixation of *-rare* takes place in connection with the reintroduction of *neko* as a (potential) subject, as represented by the dotted line. To put it another way, in (50) *-rare* spells out the head that promotes the patient *nezumi*, whereas in (49), it spells out the head that promotes the agent *neko* 'cat'. Importantly, in (49), the affixation of *-rare* does not take place before *neko* is promoted because, with both

[25] Passive-of-causative sentences have had various names including "causative passive" in Tsujimura (1996), who used the same label for our causative-of-passive (47), "indirect causative" in Harley (2013) and "causative-passive" in Agoyagi (2021).

neko and *nezumi* being in-situ, passivization has not taken place. Once the promotion of *neko* takes place, it represents passivization; the affixation of *-rare* thus takes place in connection with it. That is, the (external) merge of *neko* in its base position precedes the affixation of *-rare* in agent-promoting passives (passive-of-causatives) such as (49). This means that syntactic operations such as introduction of agents (*neko*), introduction of causers (*John*) and promotion of agents (*neko*) are mirrored by morphological operations, i.e., spellings of the agent-introducing head, of the causer-introducing head, and of the agent-promoting head, namely, root-realization (*tabe*), affixation of CS (*-sase*), and affixation of PS (*-rare*), respectively.

Notably, the height of the position of a passivized argument varies between (49) and (50); therefore, the height of the affixation of *-rare* varies accordingly. The affixation of *-rare* in *-sase-rare* (49) is higher than that of root-selecting *-rare* in (50). That is, passivization in (49) and that in (50) are not at the same level. However, causativization in (49) and passivization in (50) are at the same level in the sense that they take place at the same height, right after the completion of transitivizing.

In summary, syn2, syn3 and lex1 must be distinguished. Lex1 is the exponent of a semi-lexical (and also semi-functional) head, which introduces the agent. Note that lex1 can only be a CS.[26] Syn2 and syn3, the exponents of purely functional heads, can either be CSs or be PSs.

(51) Distributional illustration of the affixing order of CSs and PSs in Mongolian and Japanese[27]

	(semi-)lexical/functional		functional	
	V	lex1	syn2	syn3
Lexical causative	V	CS		
Passive of lexical causative (in both languages)	V	CS	PS	
Causative of lexical causative (in both languages)	V	CS	CS	
Passive of passive (in neither language)				
Causative of passive (in both languages)	V	CS	PS	CS
Passive of syntactic causative (in Japanese)	V	CS	CS	PS
Causative of syntactic causative (in Mongolian)	V	CS[28]	CS	CS

4 Split VoiceP

The facts about causative-passive interaction discussed in Section 3 suggest that the subject-predicate relationship arises in a layered structure with arguments introduced as potential subjects at various heights, which is mirrored by the layered affixation of causative and passive suffixes. Our first task in this section is then to identify the functional head, X, that introduces arguments as potential subjects invariably in passives and actives (including transitives and causatives).

(52) [$_{\text{XP3: causative}}$ *Ziroo* [$_{\text{XP2: passive}}$ *Hanako$_i$* [$_{\text{XP1: active/transitive}}$ *Taroo* [$_{\text{VP}}$ t_i *sikar*] φ] ***-are***] ***-sase***]

[26] Lex1 can also be PS_{lex}, which, however, is hardly interpreted as passive and does not concern us for our present purposes. See Section 2 for relevant discussion.

[27] I take it that lexical causatives do not differ structurally from simple transitives in the sense that both introduce EA (However, see Harley 2013 for a different position). Morphologically, however, they differ in that lexical causatives involve CS while simple transitives normally do not.

[28] CS here is either present or absent. When it is present, a longer string lex1-syn2-syn3 is obtained. See Janhunen (2012: 149) for relevant discussion on Mongolian examples and Aoyagi (2021: 101) for discussion on Japanese.

In actives, following Kratzer (1996), X is Voice (roughly corresponding to Chomsky's (1995) v). The Kratzerian Voice is not designated for a passive subject but for an external argument, EA. EA has three important properties. It is the highest one among the DPs within the transitive structure, being assigned an external role, agent or experience, etc.; it is merged outside VP, bearing no thematic relationship with the root verb; it is a potential subject (sbj). It is crucial to note here that EA is a sbj because it is the highest DP but not because it is EA; that is, it is a sbj because of its syntactic position but not because of its thematic role. If a DP occurs higher than EA, then that DP is a sbj, although it is not EA. Therefore, being EA is not a necessary condition for being a sbj, although it may be a sufficient condition. It is thus the relative height of the position of a DP, not the role it bears, that determines whether it is a sbj. In creating a subject-predicate relationship, VP (or any extended projection of it) requires a DP, new or old, to occur outside it, regardless of its role. Merging a new DP into the derivation in addition to an old DP yields a two-place structure, which we call "transitive". Strictly speaking, a structure (or verb) is transitive because it contains a sbj, not because it contains EA. What is required by structure extension is introduction of a sbj (and/or a subject-predicate relationship) rather than an external role.

Importantly, sbjs are introduced by syntactic heads of the same substance. It is not the case that sbj1 is introduced by X (v, for example) while sbj2 by Y (T, for example). It must be the case that if sbj1 is introduced by X1, sbj2 is introduced by X2, X1 and X2 being heads of the same substance. Therefore, if Voice (X1) introduces EA (e.g., *Taroo*) in actives, a head of the same substance as Voice (X2) must be able to introduce IA (e.g., *Hanako*) in passives because both EA and IA are introduced as sbjs.[29] Accordingly, that head (X2) is another Voice. This said, the head-argument correlation is in fact a Voice-sbj correlation and arguments are sbjs upon being introduced by Voice heads. It is thus reasonable that X1, X2 and X3 are all Voice heads, which are split out of VoiceP, as represented below.

(53) Split VoiceP: $[_{\text{VoiceP3}}$ DP $[_{\text{VoiceP2}}$ DP $[_{\text{VoiceP1}}$ DP $[_{\text{VP}}$ DP $]]]]$

Thus, the surface subject of passives as well as that of actives is (re)introduced beforehand by one of the separate heads of the Split VoiceP to get assigned subjecthood, thereby creating a subject-predicate relationship.

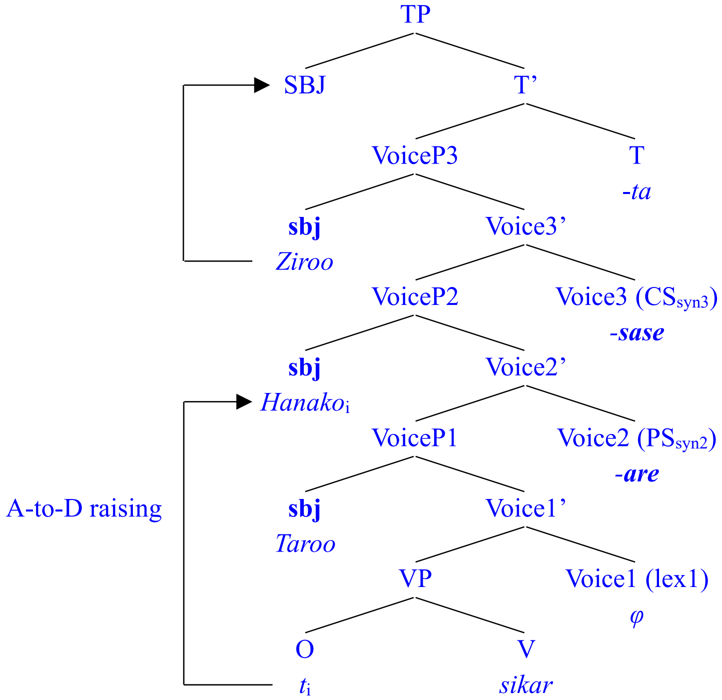
Given that the A-to-D raising of IA in causative-of-passives instantiates A-movement, it is not deniable that a passivizing head (i.e., X2 or Voice2) can introduce an argument. Importantly, this yields no difference between passives and transitives with respect to the relevant heads' ability to introduce arguments. Given that the passive is the voice proper and that passivization is successive-cyclic, it is true that passives are derived by internally merging (moving) IA into Spec of a Voice head. Consequently, EA and IA are both introduced by Voice, one via EM, the other via IM.

In what follows, let us derive the voice constructions noted in Section 3 over the proposed Split VoiceP structure. We first look at the derivation of causative-of-passive sentences, which involve A-to-D raising.

[29] IA is already assigned a patient role within VP and it will not be assigned another role when it is reintroduced (internally merged) outside VP.

(54) Structure of causative-of-passive sentence, PS_{syn}-CS_{syn} (47):

[$_{TP}$ *Ziroo*$_j$*-ga* [$_{VoiceP3}$ t_j [$_{VoiceP2}$ *Hanako*$_i$*-ni* [$_{VoiceP1}$ *Taroo-ni* [$_{VP}$ t_i *sikar*] *φ*] ***-are***] ***-sase***]*-ta*]



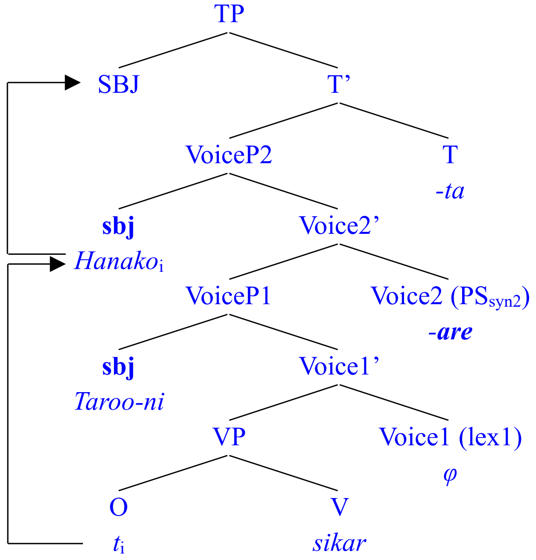
Three arguments, *Taroo* (EA), *Hanako* (IA), and *Ziroo* (causer), are (re)introduced as sbj by Voice1, Voice2, and Voice3, respectively. A-to-D raising of *Hanako*, which is base-generated within vP, represents passivization. Importantly, the last-merged sbj *Ziroo*, the highest argument, is promoted to the surface nominative subject (SBJ) in Spec of TP and assigned NOM. *Taroo* and *Hanako*, which are also sbjs, end up with non-NOM since they are not last-merged. The case of a non-last-merged sbj depends on the case marking system of a given language. In Japanese, DAT is assigned to the passivized IA, whereas ACC is assigned in Mongolian.[30]

However, when VoiceP2 does not extend to VoiceP3, *Hanako* is a last-merged sbj and promoted to SBJ, as in (55). The same holds true of *Taroo* if VoiceP1 is not selected by Voice2.

[30] ACC is also assigned alternatively in Japanese. The differential case-marking in this case has to with the semantics of causatives such as affectedness, volition, animacy, and so on. The same is true of non-passivized causees. In Japanese, again DAT and ACC alternate for non-passivized causees while in Mongolian, INS and ACC alternate. Which of the two alternants to use is dependent on the parametric variety of causative semantics (Dixon 2000).

(55) Passive-of-transitive (and passive-of-lexical-causative) (46):

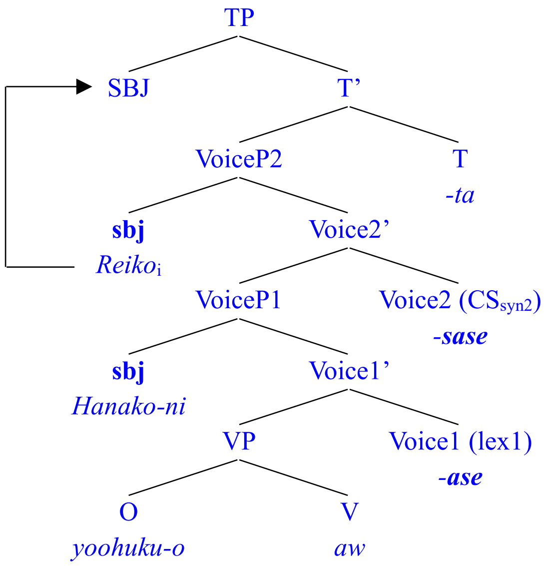
[$_{\text{TP}}$ *Hanako$_i$-ga* [$_{\text{VoiceP2}}$ t_i [$_{\text{VoiceP1}}$ *Taroo-ni* [$_{\text{VP}}$ t_i *sikar*] φ] ***-are***] *-ta*]



Note that Voice is not endowed with any dedicated feature, such as [PASSIVE], [ACTIVE] and [CAUSE], contrary to some claims in the literature, which have often postulated such features for capturing voice alternance. Thus, none of the split heads Voice1, Voice2, and Voice3 is predetermined as something such as a passive Voice or an active Voice. Causative and passive semantics is determined configurationally. Specifically, it is determined by the type of argument (IA or EA) and the type and height of the merge (IM or EM) of them. Therefore, Voice2 and Voice3 may either be passivizing heads, spelled out by PS, or be causativizing heads, spelled out by CS. When Voice2 introduces a new DP as a sbj via EM, causative semantics is produced.

(56) Causative-of-lexical-causative (40):

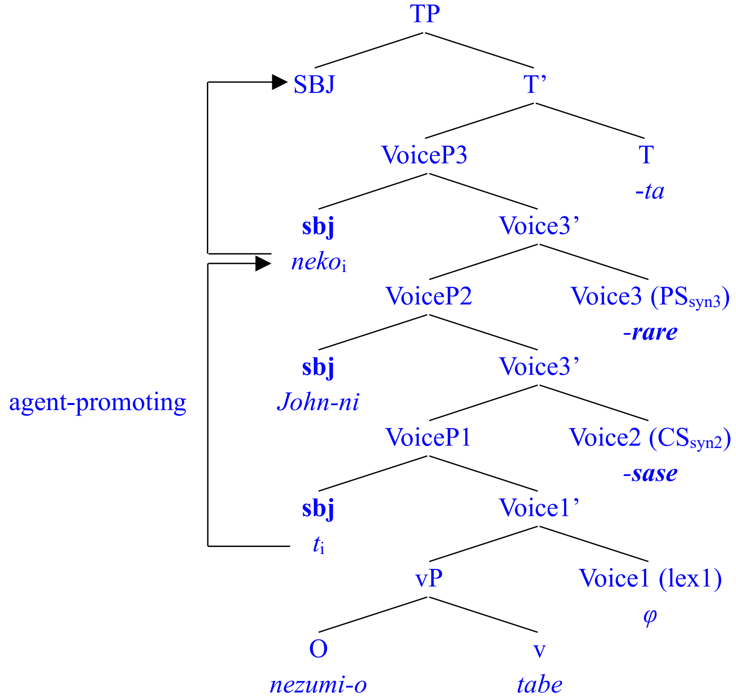
[$_{\text{TP}}$ *Reiko$_i$-ga* [$_{\text{VoiceP2}}$ t_i [$_{\text{VoiceP1}}$ *Hanako-ni* [$_{\text{VP}}$ *yoohuku aw*] ***-ase***] ***-sase***] *-ta*]



When VoiceP2 is selected by Voice3, which introduces a DP as a sbj via IM, passive-of-causative semantics is assigned to the structure.

(57) Passive-of-syntactic-causative (49):

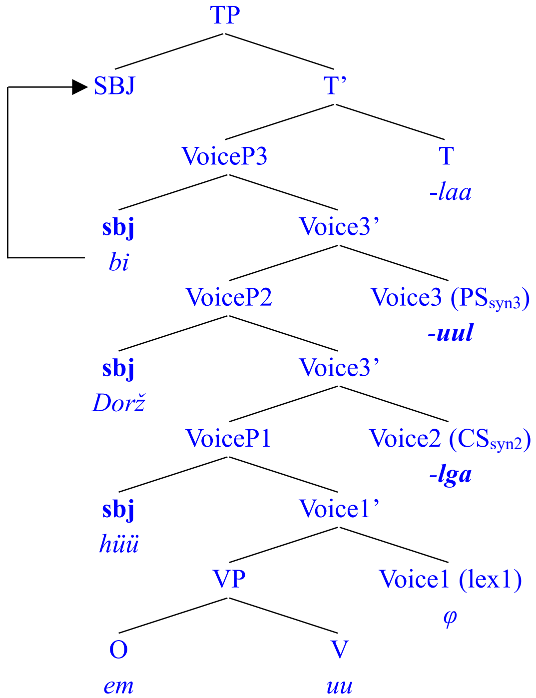
[$_{\text{TP}}$ *neko$_i$-ga* [$_{\text{VoiceP3}}$ t_i [$_{\text{VoiceP2}}$ *John-ni* [$_{\text{VoiceP1}}$ t_i [$_{\text{VP}}$ *nezumi tabe*] *φ*] ***-sase***] ***-rare***] *-ta*]



If Voice3 introduces a new DP as a sbj via EM, then causative-of-causative semantics is assigned.

(58) Causative-of-syntactic-causative (45):

[$_{\text{TP}}$ *bi$_i$* [$_{\text{VoiceP3}}$ t_i [$_{\text{VoiceP2}}$ *Dorž-oor* [$_{\text{VoiceP1}}$ *hüü-d* [$_{\text{VP}}$ *em-ig uu*] *φ*] ***-lga***] ***-uul***] *-laa*]



Voice1 cannot be a passivizing head because the DP introduced into its Spec does not move over some DP else. If Voice1 introduces a DP as a sbj via EM, the yielded structure is transitive (or lexical causative), and if it introduces a DP as a sbj via IM, a one-place structure such as an unaccusative or a middle is yielded.[31]

5 Conclusion and discussion

This paper has reported certain facts about causative-passive interactions in Japanese and Mongolian. Five patterns of interactions were discussed in connection with Mirror Principle, where a successive morphological affixation mirrors consecutive syntactic operations including causativization and passivization at different levels. It was argued that the affixation of a particular kind of voice morpheme takes place in connection with the introduction of a particular kind of argument, whether it is internal or external.

On that basis, this paper, from a formal perspective, has attempted to justify that Voice is a sbj-introducing head and that passivization, when it involves NOM, is not one-step movement but rather successive-cyclic movement. That is, passive subjects stop over an intermediate position before reaching their surface NOM position.[32] This is compatible with Legate's (2003) observation that VP-internal elements move upward in a successive-cyclic manner. All this led to the claim that VoiceP splits into separate Voice projections, where the head of each projection introduces an argument as a potential subject (sbj) via either IM or EM and is spelled out as a causative or passive morpheme; a last-merged sbj is promoted to Spec of TP to become the surface subject (SBJ), with others, if any, ending up with non-nominative (dative, instrumental, accusative, etc) case.

From the proposed analysis, it follows that introduction of arguments, regardless of its thematic role, is allowed by Free Merge (Chomsky 2013, 2015) — nothing prevents a DP from (re)merging into the derivation from the numeration or from a constructed syntactic object.

An important implication that the proposed Split VoiceP approach carries for the Voice theory is that Voice is the very argument-introducer (sbj-introducer, in the stricter sense), which introduces either EA or IA as a sbj. In this sense, the subject of passives is introduced by Voice via IM and there does not exist an expletive variant of Voice as a syntactic head, contrary to some existing claims. The table in (59) summarizes the various characterizations of Voice as a syntactic head available in the literature.

(59) Comparisons and correspondences among labels for (non-)argument-introducers

<table>
<tr><th></th><th>caus-of-pass</th><th>pass-of-caus</th><th>syn causative</th><th>syn passive</th><th>lex causative</th><th>transitive</th></tr>
<tr><td>the mainstream</td><td></td><td></td><td></td><td></td><td colspan="2">v</td></tr>
<tr><td>Kratzer (1996)</td><td></td><td></td><td></td><td></td><td colspan="2">Voice[+D]</td></tr>
<tr><td>Collins (2005, 2024)</td><td></td><td></td><td></td><td>Voice</td><td colspan="2">v</td></tr>
<tr><td>Pylkkänen (2008)</td><td></td><td>Voice[–D]</td><td>Voice[+D]</td><td>Voice[–D]</td><td colspan="2">v_{cause} / Voice[+D]</td></tr>
<tr><td>Bruening (2012)</td><td></td><td></td><td></td><td>Voice[–D]</td><td colspan="2">Voice[+D]</td></tr>
<tr><td>Harley (2013)</td><td></td><td>Voice[–D]</td><td>Voice[+D]</td><td>Voice[–D]</td><td>v</td><td>Voice[+D]</td></tr>
<tr><td>Alexiadou et al. (2015)</td><td></td><td></td><td></td><td>Voice[–D]</td><td>Voice_{causer}[+D]</td><td>$\text{Voice}_{agent/holder}$[+D]</td></tr>
<tr><td>Legate et al. (2020)</td><td></td><td></td><td></td><td>Voice[–D]</td><td colspan="2">Voice[+D]</td></tr>
<tr><td>Nie (2020)</td><td></td><td></td><td>Voice[+D]</td><td></td><td>Voice[+D]</td><td></td></tr>
<tr><td>this paper</td><td colspan="2">Voice3</td><td colspan="2">Voice2</td><td colspan="2">Voice1</td></tr>
</table>

[31] See Bai and Xia (2025) for discussion of middles and unaccusatives.
[32] See also Bai (2025)

This table suffices to show the overuse or even misuse of Voice, which, an undesirable result, was led to by the postulation of Voice[–D] as an expletive variant of the Voice head. The Voice[–D] approach holds the following views. The active-passive distinction is attributed to the difference in selectional ability between heads, which have ultimately been assumed to be distinctive features of Voice. If a Voice head can introduce a DP as its argument, then it is often said to be Voice[+D] or Voice[act], and if a voice head lacks this ability, then it is said to be Voice[–D] or Voice[passive/non-act]. There have, therefore, often been assumed to exist two opposite variants of Voice, one for actives (including transitives/causatives) and another for passives, middles, and sometimes unaccusatives. Under a Voice[–D] approach, UG provides different types of Voice heads in the lexicon and therefore various voice types such as causative/active and passive are determined by Voice heads; that is, the meanings interpreted for causative/active and passive as voice constructions are predetermined in the lexicon.[33] This would allow one to say that, for example, a sentence is a passive sentence because it is derived via Voice with a passive feature. This would amount to saying that a sentence is a passive sentence because its voice is passive, a sentence is a middle sentence because its voice is middle, and a sentence is an active sentence because its voice is active. However, one, whether generativist or not, would not be impressed by this (see Haider 2024: 14). An important consequence of the Voice[–D] theory is that the subject of passives has to do with neither Voice[–D], the passive head postulated by it, nor the passive semantics syntactically.[34] In contrast, the Split VoiceP analysis advocated in this paper holds that UG provides Voice[+D], not Voice[–D], in the inventory of functional heads.

It is hoped that this paper has provided a unified account of voice alterations and simplified the clause-formation mode: Clauses are built by consecutively introducing arguments as potential subjects in Spec of separate Voice heads. It is also hoped that the facts about the passive-causative interaction discussed in this paper will serve as an empirical basis in conducting a more accurate characterization of Voice as a universal syntactic category and in articulating the architectural status of the voice domain.

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[33] Alexiadou et al. (2018: 419), for example, states, "The lexicon can provide two variants of the semi-functional Voice-head, which differ in their syntax and semantics".

[34] While the passive as a voice construction has lacked a consistent definition, this paper takes a DP to be a passive subject if it satisfies (i) below, which spells out the defining property of the construction.

(i) If DP instantiates DP(λy (P y, x, t_y)), where y is lower than x, x being non-nominative, in the thematic hierarchy with respect to the predicate P, then P is passive and DP is subject of P.

"Passive semantics" then refers to the meaning interpreted for such a configuration. On a Voice[–D] view, it would be the case that what solely contributes to producing the passive meaning is the agent-suppressing property of the Voice[–D] head, or a dedicated voice-specifying feature like [passive/non-act] on it. That is, syntax is not truly autonomous in producing the passive meaning on a Voice[–D] view, which would not assert a configuration such as "(P y, x, t_y)", appealing only to non-nominative y, for deriving passives.

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Appendix: Gloss alignment to the examples using Japanese and Mongolian characters

No.	Example
(4)	*Taroo-ga Hiroko-ni pizza-o tabe-**sase**-ta.* 太郎が弘子にピザを食べさせた。
(5/36)	*Hiroko-ga pizza-o tabe-**sase**-**rare**-ta.* 弘子がピザを食べさせられた。
(6/46)	*Hanako-ga Taroo-ni sikar-**are**-ta.* 花子が太郎に叱れた。
(7/37/47)	*Ziroo-ga Hanako-o/ni Taroo-ni sikar-**are**-**sase**-ta.* 次郎が花子を/に太郎に叱られさせた。
(11)	*Itgegčid bolon tede-ne ariun sudaruud gal-d šat-**aa**-**gd**-b.* [illegible] Итгэгчид болон тэдний аригун содрууд галд шатаагдав.
(12)	*End olan üildber bai-**guul**-**gd**-ž baina.* [illegible]

	Энд олон үйлдвэр байгуулагдаж байна.
(13)	*Hairlah sedgel ni uul yabdal-ig dahin dahin bod-**gd**-**uul**-na.* ᠬᠠᠶᠢᠷᠯᠠᠬᠤ ᠰᠡᠳᠬᠢᠯ ᠨᠢ ᠤᠤᠯ ᠶᠠᠪᠤᠳᠠᠯ ᠢ ᠳᠠᠬᠢᠨ ᠳᠠᠬᠢᠨ ᠪᠣᠳᠤᠭᠳᠠᠭᠤᠯᠤᠨ᠎ᠠ ᠃ Хайрлах седгел нь уул явдалыг дахин дахин бодогдуулна.
(14)	*Dorž-ig aab-d-ni tani-**gd**-**uul**-h-güi-in tuld sahal naa-san.* ᠳᠣᠷᠵᠢ ᠢ ᠠᠪᠤᠭᠠᠳ ᠨᠢ ᠲᠠᠨᠢᠭᠳᠠᠭᠤᠯᠬᠤ ᠶᠢᠨ ᠲᠤᠯᠠᠳᠠ ᠰᠠᠬᠠᠯ ᠨᠠᠭᠠᠰᠠᠨ ᠃ Доржийг аавд нь танигдуулахгүйн тулд сахал наасан.
(15)	*Bi düü-geer hubčas-aa hat-**aa**-**lga**-san.* ᠪᠢ ᠳᠡᠭᠦᠦ ᠪᠡᠷ ᠬᠤᠪᠴᠠᠰᠤ ᠪᠠᠨ ᠬᠠᠲᠠᠭᠠᠯᠭᠠᠰᠠᠨ ᠃ Би дүүгээр хувцсаа хатаалгасан.
(16)	*Bagš nad-aar neg huudas züil orč-**uul**-**uul**-b.* ᠪᠠᠭᠰᠢ ᠨᠠᠳᠠ ᠪᠠᠷ ᠨᠢᠭᠡ ᠬᠤᠭᠤᠳᠠᠰᠤ ᠵᠦᠢᠯ ᠣᠷᠴᠢᠭᠤᠯᠤᠭᠤᠯᠪᠠ ᠃ Багш надаар нэг хуудас зүйл орчуулуулав.
(17)	*Hutgan hürd zööltei yih zaan č usan-d ab-**ta**-**gd**-b.* ᠬᠤᠲᠤᠭ᠎ᠠ ᠬᠦᠷᠳᠦ ᠵᠡᠭᠦᠯᠲᠡᠢ ᠶᠡᠬᠡ ᠵᠠᠭᠠᠨ ᠴᠤ ᠤᠰᠤᠨ ᠳᠤ ᠠᠪᠲᠠᠭᠳᠠᠪᠠ ᠃ Хутган хүрд зүүлттэй их заан ч усанд автагдав.
(19)	*Bi gar-aa nohai-d haž-**uul**-san.* ᠪᠢ ᠭᠠᠷ ᠢᠶᠠᠨ ᠨᠣᠬᠠᠢ ᠳᠤ ᠬᠠᠵᠠᠭᠤᠯᠤᠭᠰᠠᠨ ᠃ Би гараа нохойд хазуулсан.
(20)	*Bi nohai-d haž-**uul**-san.* ᠪᠢ ᠨᠣᠬᠠᠢ ᠳᠤ ᠬᠠᠵᠠᠭᠤᠯᠤᠭᠰᠠᠨ ᠃ Би нохойд хазуулсан.
(21)	*Bi nohai-d haž-**gd**-san.* ᠪᠢ ᠨᠣᠬᠠᠢ ᠳᠤ ᠬᠠᠵᠠᠭᠳᠠᠭᠰᠠᠨ ᠃ Би нохойд хазагдсан.
(22)	*Nad-d neg sonin duu sons-**d**-b.* ᠨᠠᠳᠠᠳᠤ ᠨᠢᠭᠡ ᠰᠣᠨᠢᠨ ᠳᠠᠭᠤ ᠰᠣᠨᠣᠰᠳᠠᠪᠠ ᠃ Надад нэг сонин дуу сонсдов.
(23)	*Nad-d hüč neme-**gd**-b.* ᠨᠠᠳᠠᠳᠤ ᠬᠦᠴᠦ ᠨᠡᠮᠡᠭᠳᠡᠪᠡ ᠃ Надад хүч нэмэгдэв.
(32)	*Bi Dorž-oor hüü-d-ee em-ig-ni uu-**lga**-**uul**-laa.* ᠪᠢ ᠳᠣᠷᠵᠢ ᠪᠠᠷ ᠬᠦᠭᠦᠳᠡ ᠪᠡᠨ ᠡᠮ ᠢ ᠨᠢ ᠤᠤᠭᠤᠯᠭᠠᠭᠤᠯᠤᠯ᠎ᠠ ᠃ Би доржоор хүүдээ эмийг нь уулгуулллаа.
(25)	*Bi dahinta Baatar-ig ir-**üül**-b.* ᠪᠢ ᠳᠠᠬᠢᠨᠲᠠ ᠪᠠᠭᠠᠲᠤᠷ ᠢ ᠢᠷᠡᠭᠦᠯᠪᠡ ᠃ Би дахинтаа баатарыг ирүүлэв.
(26)	*Bi Baatar-ig dahinta ir-**üül**-b.* ᠪᠢ ᠪᠠᠭᠠᠲᠤᠷ ᠢ ᠳᠠᠬᠢᠨᠲᠠ ᠢᠷᠡᠭᠦᠯᠪᠡ ᠃ Би баатарыг дахинтаа ирүүлэв.
(27)	*Bid daisan-ig zoriud sön-**öö**-b.* ᠪᠢᠳᠡ ᠳᠠᠶᠢᠰᠤᠨ ᠢ ᠵᠣᠷᠢᠭᠤᠳ ᠰᠥᠨᠥᠭᠡᠪᠡ ᠃ Бид дайсныг зориуд сөнөөв.

(35)	*Hoteru-no mae-de or-**os**-**are**-ta.* ホテルの前で降ろされた。
(38)	*Musuko-o seken-no aranami-ni mom-**are**-**sase**-ru.* 息子を世間の荒波に揉まれさせる。
(39)	*Taroo-ga Jiroo-ni sensei-o nak-**as**-**ase**-ta.* 太郎が次郎に先生を泣かさせた。
(40)	*Reiko-ga Hanako-ni yoofuku-o aw-**ase**-**sase**-ta.* 玲子が花子に洋服を合わさせた。
(44)	*Dorž aab-d-aa tani-**gd**-san.* ᠳᠣᠷᠵᠢ ᠠᠪᠤ ᠳᠤ ᠪᠠᠨ ᠲᠠᠨᠢᠭᠳᠠᠭᠰᠠᠨ ᠃ Дорж аавдаа танигдсан.
(48)	*Ziroo-ga Taroo-ni Hanako-o sikar-**ase**-ta.* 次郎が太郎に花子を叱らせた。
(49)	*Neko-ga John-ni nezumi-o tabe-**sase**-**rare**-ta.* 猫がジョンに鼠を食べさせられた。
(50)	*Nezumi-ga neko-ni tabe-**rare**-ta.* 鼠が猫に食べられた。
(60)	*Taroo-ga Hanako-ni piano-o asa-made hik-**are**-ta.* 太郎が花子にピアノを朝まで引かれた。
(61)	*Taroo-ga coochi-ni nak-**are**-ta.* 太郎がコーチに泣かれた。

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